SEC:: SR. JEE MAINS APEX | GTM-25 | Date: 24.01.2025
Time: 3.00Hrs. | JEE MAINS -2025 | Max. Marks: 300 M

KEYSHEET

PHYSICS

1	2	3	4	5	6	7	8	9	10
C	B	C	C	C	A	C	B	D	C
11	12	13	14	15	16	17	18	19	20
A	C	A	D	B	D	D	C	A	D
21	22	23	24	25					
4	2	15	3	2					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
C	C	A	A	C	A	B	B	A	B
36	37	38	39	40	41	42	43	44	45
B	B	D	D	A	B	C	A	D	C
46	47	48	49	50					
32	5	9	4	4					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
B	C	A	C	C	C	A	B	A	A
61	62	63	64	65	66	67	68	69	70
D	D	C	D	B	A	C	D	B	A
71	72	73	74	75					
42	5	0	324	7					

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PHYSICS

1. Acceleration of the system and hence tension (T) on the string attached with the pulley is same in both the cases. Hence

$$F_1 = F_2 = 2T$$

2. $$V_T \propto r^2\rho \Rightarrow V_T \propto r^2 \frac{m}{r^3} \Rightarrow V_T \propto \frac{1}{r}$$

3. $$F = \frac{K(Q)(2Q)}{r^2}$$

When allowed to touch, we have

$$F' = \frac{K\left(\frac{Q}{2}\right)\left(\frac{Q}{2}\right)}{r^2}$$

$$\Rightarrow F' = \frac{F}{8}$$

4. $$\upsilon = \sqrt{\frac{\gamma p}{\rho}} = \sqrt{\gamma\left(\frac{RT}{M}\right)}$$

$$\Rightarrow \frac{\upsilon_{H_2}}{\upsilon_{O_2}} = \sqrt{\frac{M_{O_2}}{M_{H_2}}} = \sqrt{\frac{32}{2}}$$

$$\Rightarrow \frac{\upsilon_{H_2}}{\upsilon_{O_2}} = 4$$

5. No force on PQ and SR.

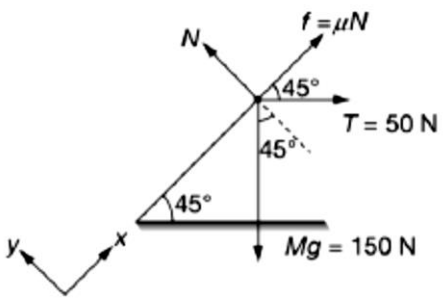
Force on PS or QR = Bil

$$= 2\times10^{-2}\times2\times0.25N = 10^{-2}N$$

$$\tau = 10^{-2}N\times0.4m = 4\times10^{-3}Nm$$

6. The string is under tension, hence there is limiting friction between the block and the plane.

Drawing free body diagram of the block.



$$\Sigma F_X = 0$$

$$\Rightarrow \quad \mu N + 50\cos 45^\circ = 150\sin 45^\circ \qquad \ldots\ldots\ldots (1)$$

$$\Sigma F_y = 0$$

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$\Rightarrow \quad N = 50\sin 45^{o} + 150\cos 45^{o}$ ……. (2)

Solving (1) and (2) we get $\mu = \frac{1}{2}$.

7. $\frac{1}{P_{eq}} = \frac{1}{P_1} + \frac{1}{P_2} + ... + \frac{1}{P_9}$

$P_{eq} = \frac{P}{9}$

8. $R = R_0 A^{1/3}$

9. $E = 2 \times \frac{5}{2} RT + 4 \times \frac{3}{2} RT = 11RT$

10. Conceptual

11. $\frac{E_0}{B_0} = V \Rightarrow \frac{E_0}{B_0} = \frac{\omega}{k}$

12. Conceptual

13. We have $\frac{R}{55} = \frac{80}{20}$

$\therefore \quad R = 220\Omega$

14. Since $T \propto l^{\frac{1}{2}}$, so we have $\frac{\Delta T}{T} = \frac{1}{2}\left(\frac{\Delta l}{l}\right) = 1\%$

$\Rightarrow \quad \frac{T' - T}{T} = \frac{1}{100}$

$\Rightarrow \quad T' = T + 0.01T = 1.01T$

15. Conceptual

16. $Y = \overline{A.(A+B)} = \overline{A} + \overline{(A+B)}$

$\Rightarrow Y = \overline{A} + \overline{A}.\overline{B} = \overline{A}(1+\overline{B})$

$\Rightarrow Y = \overline{A}$

17. Since, we have

$i = -\sqrt{2}\cos\left(\omega t + \frac{\pi}{4}\right)$

$\Rightarrow \quad i = \sqrt{2}\sin\left(\omega t + \frac{\pi}{4} - \frac{\pi}{2}\right)$

Hence, phase difference between V and i *is* $\frac{\pi}{2}$. So, power consumed is zero.

18. Conceptual

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19. Area under P-x graphs $= \int pdx = \int \left(m\frac{dv}{dt} \right) vdx$

$$= \int_1^v mv^2 dv = \left[\frac{mv^3}{3} \right]_1^v = \frac{10}{7\times 3}\left(v^3 - 1 \right)$$

From graph: area $= \frac{1}{2}(2+4)\times 10 = 30$

$$\therefore \frac{10}{7\times 3}\left(v^3 - 1 \right) = 30 \qquad \therefore v = 4m/s$$

20. $\vec{\Delta v} = \vec{v}_2 - \vec{v}_1$

$$\left| \vec{\Delta v} \right| = \sqrt{v^2 + v^2 - 2v^2 \cos\left(40^\circ \right)}$$

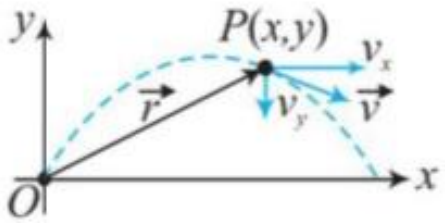
$$\Rightarrow \left| \vec{\Delta v} \right| = v\sqrt{2 - 2\cos\left(40^\circ \right)}$$

$$\Rightarrow \left| \vec{\Delta v} \right| = \sqrt{2}v\sqrt{1 - \cos\left(40^\circ \right)}$$

$$\Rightarrow \left| \vec{\Delta v} \right| = \sqrt{2}v\sqrt{2\sin^2\left(20^\circ \right)} \qquad \left\{ \text{Q} 1 - \cos\left(2\theta \right) = 2\sin^2\theta \right\}$$

$$\Rightarrow \left| \vec{\Delta v} \right| = 2v\sin\left(20^\circ \right)$$

21. In this case the angular momentum of the particle will not remain constant as there is always torque of gravity is acting . Let us take the origin at *O*, *x*-axis along the horizontal and *y*-axis along the vertically upward direction as shown in figure. For horizontal motion during the time0 to *t*,



$$v_x = v_0\cos 45^\circ = \frac{v_0}{\sqrt{2}} \text{ and } x = v_x t = \frac{v_0}{\sqrt{2}}\frac{v_0}{g} = \frac{v_0^2}{\sqrt{2}g}$$

For vertical motion, using $v_y = v_{oy} - gt$, we get

$$v_y = v_o\sin 45^\circ - g\left(\frac{v_0}{g} \right) = \frac{\left(1 - \sqrt{2} \right)}{\sqrt{2}}v_0 \text{ and } y = \left(v_o\sin 45^\circ \right)t - \frac{1}{2}gt^2$$

$$\Rightarrow y = \frac{v_0}{\sqrt{2}}\left(\frac{v_0}{g} \right) - \frac{1}{2}g\left(\frac{v_0}{g} \right)^2 = \frac{v_0^2}{2g}\left(\sqrt{2} - 1 \right)$$

The angular momentum of the particle at time *t* about the origin is

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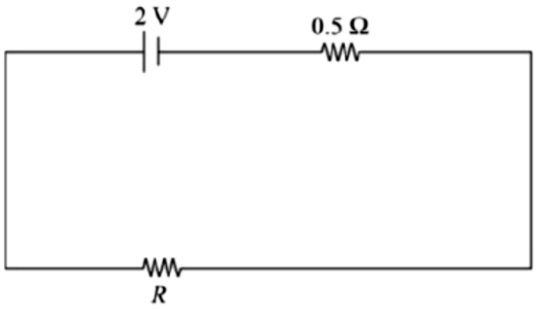
$$L=\vec{r}\times\vec{p}=m\left(\vec{r}\times\vec{v}\right)$$

$$\left(x\hat{i}+y\hat{j}\right)\times\left(v_x\hat{i}+v_y\hat{j}\right)=m\left(xv_y\hat{k}-yv_x\hat{k}\right)$$

$$m\left[\left(\frac{v_0^2}{\sqrt{2}g}\right)\frac{v_0}{\sqrt{2}}\left(1-\sqrt{2}\right)-\frac{v_0^2}{2g}\left(\sqrt{2}-1\right)\frac{v_0}{\sqrt{2}}\right]\left(\hat{k}\right)$$

$$=\frac{mv_0^3}{2\sqrt{2}g}\left(-\hat{k}\right)$$

22. Considering the two batteries in parallel equivalent circuit is shown in figure below.



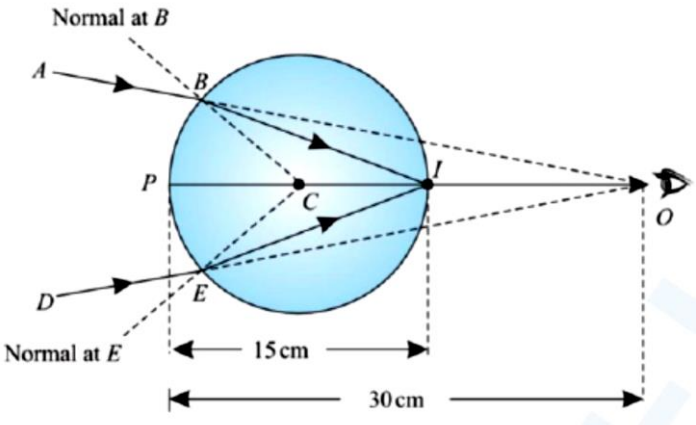
Maximum power across R is obtained when $R=r=0.5\Omega$

Maximum current in circuit is given as

$$i=\frac{2}{R+r}=2A$$

$$\Rightarrow P_{\max}=i^2R=(2)^2(0.5)=2W$$

23. Let the incident rays on the sphere from left. In the absence of the spherical surface, they converge at *O*, 30 cm from the pole *P*. Thus, *O* serves as the virtual object from first refraction at left surface,



Here, $\mu=PO=+30cm$, $R=+\frac{15}{2}cm$, $\mu_1=1$ and $\mu_2=\frac{3}{2}$

Let the distance of the image *l* from *P* be *v*.

Using, $\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$

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or $\frac{3/2}{v}-\frac{1}{(+30)}=\frac{(3/2-1)}{(+15/2)}\Rightarrow v=+15cm$

Thus, the image formed is real.

24. $B=\frac{\mu_0 i}{2r}-\frac{\mu_0 i}{2(2r)}+\frac{\mu_0 i}{2(2^2 r)}+....+\infty$

$=\frac{\mu_0 i}{2r}\left[1-\frac{1}{2}+\frac{1}{2^2}-\frac{1}{2^3}+...+\infty\right]$

$=\ =\frac{\mu_0 i}{2r}\left[\frac{1}{1-\left(-\frac{1}{2}\right)}\right]=\frac{\mu_0 i}{3r}$

25. $V_L=L\frac{di}{dt}\Rightarrow di=V_L dt$

$L(i_f-i_0)=$ Area under (V_L-t) graph.

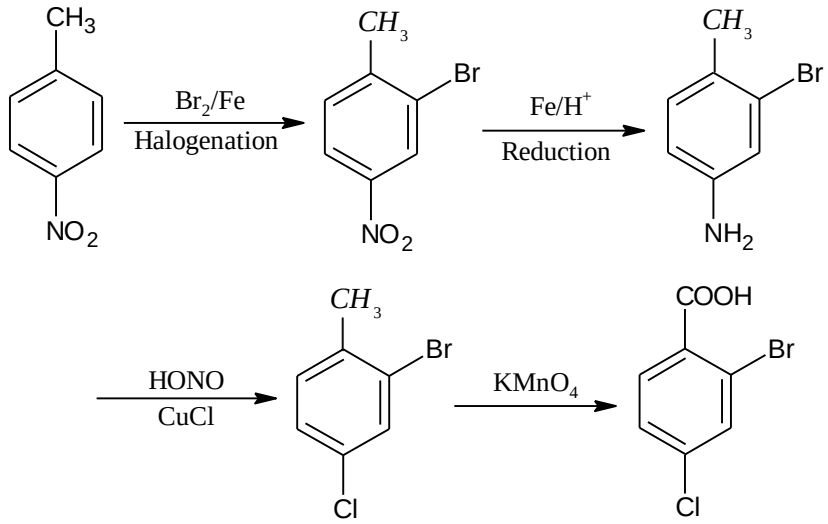
$4(i_f-0)=\frac{1}{2}(8)(2) \quad \Rightarrow i_f=2A$

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26. $\left[Co(CO)_4\right]^+$ diamagnetic, dsp^2 (square planner)

27. $6d^1$

28.



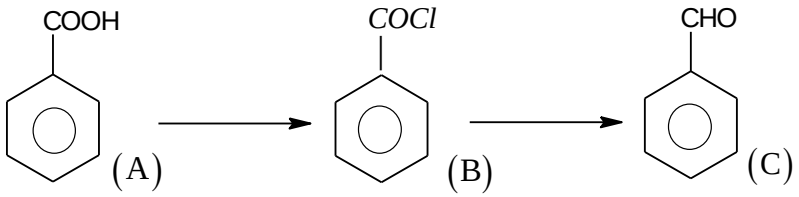
29. Conceptual

30. Conceptual

31. Conceptual

32. Conceptual

33.



Rosenmund's Reduction

(B)⟶(C)

34. Statements given in A, C and D are correct, while in B it is incorrect.

The correct form of incorrect statement is, Aniline is immiscible in water, and hence it can be purified by steam distillation.

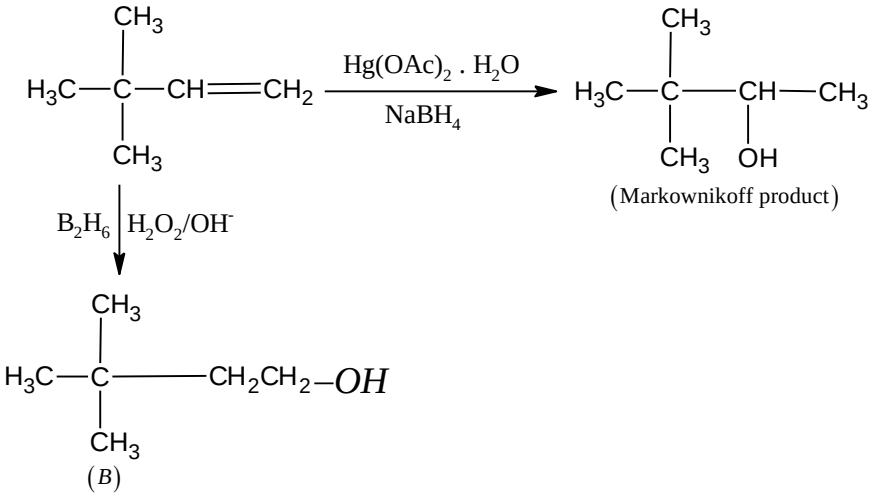
35. Both Statement I and Statement II are true.

The NH_2 group in aniline id *ortho* and *para* directing and a powerful activating group as NH_2 has strong $+M$ effect.

Thus, aniline does not undergo Friedel-Craft's reaction because it will form complex with $AlCl_3$, that will deactivate the benzene ring.

36. The reaction is represented as

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(Anti- Markownikoff product)

37. $3MnO_4^{-2} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

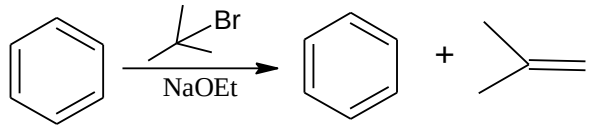
38. As trend for decreasing bond energy in the given hydrides is $H_2O > H_2S > H_2Se > H_2Te$, hence their enthalpy of formation also varies in the same trend.

As extend of delocalization of lone pair on central atom into vacant *d*-orbital increases downward hence bond angle decreases from H_2O to H_2Te.

Order of B.P. $H_2O > H_2Te > H_2Se > H_2S$

Boiling point of H_2Te, H_2Se and H_2S us decided by van der Waals' forces *i.e.*, higher van der Waals' force among molecules more will be B.P. But B.P. of H_2O is decided by H-bond. Strength of H-bond in H_2O is greater than van der Waals' forces of attraction among any other given molecular species.

39. At constant temperature, K_p remains constant. With change of pressure, *x* will change in such a way that K_p remains a constant.

40. 

'2', '3', '4' gives tert – butyl Benzene

41. Moles of $P_4 = \frac{1.24}{124} = 0.01 mol$

$P_4(s) + 3NaOH(aq) + 3H_2O(l) \longrightarrow PH_3(g) + 3NaH_2PO_2(aq)$

Reaction of PH_3 with $CuSO_4$, can written as

$3PH_3 + 3CuSO_4 \longrightarrow Cu_3P_2 + 3H_2SO_4$

2 moles of PH_3 reacts with 3 moles $CuSO_4$ by double displacement reaction.

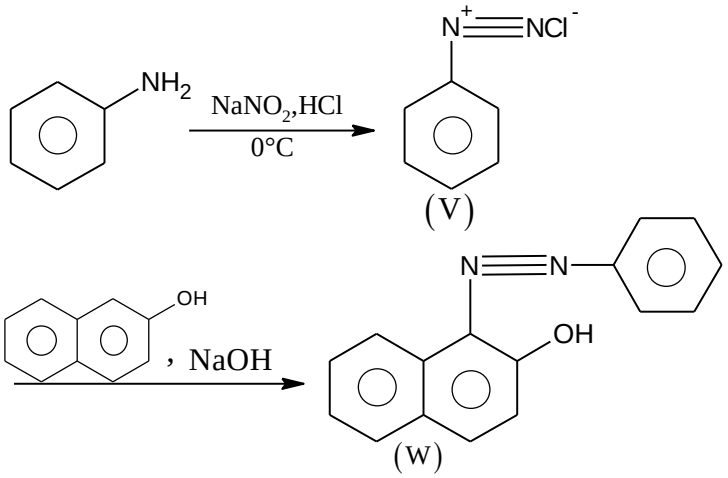
0.01 mole PH_3 reacts with $\frac{3 \times 0.01}{2} = 0.015$ mol $CuSO_4$

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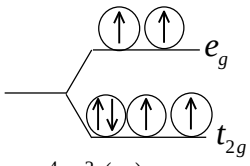
Mass of $CuSO4 = 0.015 \times 159 = 2.38\,g$

42. Conceptual

43.

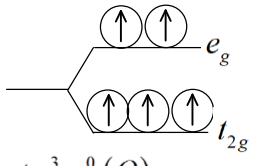


44. *i.* $\left[{}^{II}Fe(H_2O)_6\right]^{+2}$ $_{WFL}$ configuration $3d^6$



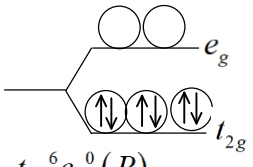
$t_{2g}{}^4 e_g{}^2 (S)$

ii. $\left[{}^{II}Mn(H_2O)_6\right]^{+2}$ $_{WFL}$ configuration $3d^5$



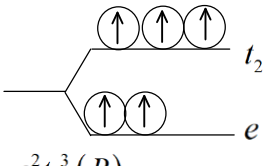
$t_{2g}{}^3 e_g{}^0 (Q)$

iii. $\left[{}^{III}Co(NH_3)_6\right]^{+3}$ $_{SFL}$ configuration $3d^6$



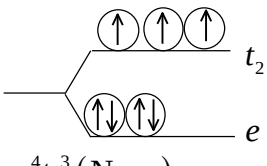
$t_{2g}{}^6 e_g{}^0 (P)$

iv. $\left[{}^{III}FeCl_4\right]^{-}$ $_{WFL}$ configuration $3d^5$



$e^2 t_2{}^3 (R)$

v. $\left[{}^{II}CoCl_4\right]^{2-}$ $_{WFL}$ configuration $3d^7$



$e^4 t_2{}^3$ (None)

45. White P Is insoluble in water but soluble in CS_2

Red P is insoluble in both water as well as CS_2

46. $3.30 \times 74.5\,g$ KCl present with 1000 g water

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Mass of solution = 1245.85 g

Volume of solution = 1038.2 mol.

Molarity $= \dfrac{3.3}{1038.2} \times 1000 = 3.18$

$= 32 \times 10^{-1}$

47. $n = \dfrac{390-180}{42} = \dfrac{210}{42} = 5$

48. $t_{1/8} = \dfrac{2.303 \log 8}{k} = \dfrac{2.303 \times 3 \log 2}{k}$

$t_{1/10} = \dfrac{2.303}{k} \log 10 = \dfrac{2.303}{k}$

$\left[\dfrac{t_{1/8}}{t_{1/10}}\right] \times 10 = \left(\dfrac{\dfrac{2.303 \times 3 \log 2}{k}}{\dfrac{2.303}{k}}\right) \times 10 = 9$

49. Only (d) is correct.

50. $2 \times 3 + 6 \times 2 + 2 \times 1 = 5 \times n_{KMnO_4}$

$n_{KMnO_4} = \dfrac{20}{5} = 4$

MATHS

51. No. of ways = ${}^{5+4-1}C_{4-1} = {}^{8}C_3$

52. $3\alpha = -8+2 = -6 \Rightarrow \alpha = -2$; $\quad 3\beta = -12+6 \Rightarrow \beta = -2$

Distance from $P(a+5,\ 2\beta+8) = P(3,\ 4)$ from $3x+4y-10=0$ is $\dfrac{|4+16-10|}{5} = 3$

53. $(Z_1 - Z_2)^2 = Z_1^2 + Z_2^2 - 2Z_1Z_2$

$16 = 4 - 6i - 2Z_1Z_2 \Rightarrow 2Z_1Z_2 = -12 - 6i \Rightarrow Z_1Z_2 = -6 - 3i$

Now, $\left(Z_1^2 + Z_2^2\right)^2 = (4-6i)^2 = Z_1^4 + Z_2^4 + 2(Z_1Z_2)^2$

$\Rightarrow 16 - 36 - 48i = Z_1^4 + Z_2^4 + 2(6+3i)^2$

$\Rightarrow Z_1^4 + Z_2^4 = -2a - 48i - 54 - 72i = -74 - 120i = a + ib$; $a - b = -24 + 120 = 46$

54. Ends of the chord are $= (5,5)$ or $(-2,-2)$

55. $x = 1,\ f(1) = e - 0 + 0 \Rightarrow f(1) = e$

Diff. w.r.t x

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$$f^1(x)=0-\left\{(x-1)\frac{1}{x}+(\ln x-1)\right\}+f(x)$$

$$\Rightarrow \frac{dy}{dx}-y=-\left\{1-\frac{1}{x}+\ln x-1\right\}=\frac{1}{x}-\ln x\ ; \qquad \frac{dy}{dx}-y=\frac{1}{x}-\ln x$$

If = $e^{-\int dx=e^{-x}}$

Solution of DE, $ye^{-x}=\int e^{-x}\left(\frac{1}{x}-\ln x\right)dx \qquad -x=t\Rightarrow dx=-dt$

$$ye^{-x}=-\int e^{t}\left(-\frac{1}{t}-\ln(-t)\right)dt$$

$$ye^{-x}=\int e^{t}\left(\ln(-t)+\frac{1}{t}\right)dt=e^{t}.\ln(+t)+C$$

$ye^{-x}=e^{-x}\ln x+C\Rightarrow y=\ln x+Ce^{x}$ at $x=1,\ y=e\ ; \quad l=0+Ce\Rightarrow C=1$

$f(x)=\ln x+e^{x}\ ;\ \ f(l)=1+e^{e}$

56. $\vec{a}=\vec{b}\times\vec{c}+2\vec{b}$

Taking dot product with $\vec{b}$

$\vec{a}.\vec{b}=2\left|\vec{b}\right|^2 \ \Rightarrow \left|\vec{a}\right|\left|\vec{b}\right|\cos\theta=2\left|\vec{b}\right|^2$

$\Rightarrow\ \cos\theta=\frac{4}{\left|\vec{a}\right|}\Rightarrow\left|\vec{a}\right|=4\Rightarrow\ \theta=0° \qquad \Rightarrow \quad \vec{a}=2\vec{b}$

Now, $\vec{b}\times\vec{c}=\vec{0}\ \Rightarrow\ \vec{b}=\vec{c}\ \ or\ \vec{b}=-\vec{c}$;

$\therefore$ Required sum $=12+8=20$

57. The directrix of the ellipse is $x=1$

Clearly the equation of common tangent is $x=1$

Equation of ellipse $P_x=ePM \quad \Rightarrow Ps^2=e^2PM^2$

$$\Rightarrow\left(x-\frac{1}{2}\right)^2+(y-1)^2=\frac{1}{4}(x-1)^2\ \Rightarrow\frac{\left(x-\frac{1}{3}\right)^2}{\frac{1}{9}}+\frac{(y-1)^2}{\frac{1}{12}}=1$$

58. Q λ lies between the roots $\qquad \therefore f(\lambda)<0$

$\Rightarrow 2\lambda^2-2(2\lambda+1)\lambda+\lambda(\lambda+1)<0$

$\Rightarrow-\lambda^2-\lambda<0 \qquad \Rightarrow\lambda^2+\lambda>0$

$\Rightarrow\lambda<-1 \qquad$ Or $\qquad \lambda>0$

$\therefore$ Least non-negative integral value of $\lambda=1$.

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59. $x-1<[x]\le x$; $2^3x-1<[2^3.x]\le 2^3.x$

$3^3.x-1<[3^3.x]\le 3^3.x$; $\oplus\dfrac{n^3.x-1<[n^3.x]\le n^3.x}{}$

$x(1^3+2^3+3^3+.....n^3)-n<[x]+[2^3.x]+[3^3.x]+.....[n^3.x]\le x(1^3+2^3+3^3+.....n^3)$

$$\lim_{n\to\infty}\frac{x\left(\frac{n(n+1)}{2}\right)^2}{n^4}=\frac{1}{n^3}<\lim_{n\to\infty}\frac{[x]+[2^3.x]+[3^3.x]+.....+[n^3.x]}{n^4}\le\lim_{n\to\infty}\frac{x\left(\frac{n(n+1)}{2}\right)^2}{n^4}$$

$\Rightarrow\dfrac{x}{4}<f(x)\le\dfrac{x}{4}\Rightarrow f(x)=\dfrac{x}{4}$; $f(f(x))=\dfrac{x}{16}$

60. Putting x=0 in given relation

$0=1-a\int_0^1 f(t)e^{-1}dt \quad\Rightarrow a\int_0^1 f(t)e^{-1}dt=1$

$\therefore\int_0^x f(t)dt=e^x-e^{2x} \quad\Rightarrow f(x)=e^x-2e^{2x}$

$f(1)+2f(2)=e-2e^2+2e^2-4e^4=e-4e^4$

61. $Mean=\dfrac{3+5+7+a+b}{5}=5\Rightarrow a+b=10$

$Variance=\dfrac{3^2+5^2+7^2+a^2+b^2}{5}-(5)^2=4$

$\Rightarrow a^2+b^2=62\ \Rightarrow(a+b)^2=2ab=62\ \Rightarrow ab=19$

So a and b are the roots of the equation $x^2-10x+19=0$

62. $f(x)=\sqrt[3]{x+\sqrt{1+x^2}}+\sqrt[3]{x-\sqrt{1+x^2}}\Rightarrow\alpha-\dfrac{1}{\alpha}=y$

$x^2-\alpha y-1=0\Rightarrow\alpha=\dfrac{y\pm\sqrt{y^2+4}}{2}$; $x+\sqrt{1+x^2}=\left\{\dfrac{y\pm\sqrt{y^2+4}}{2}\right\}^3=a$

$1+x^2=(a-x)^2\Rightarrow 1=a^2-2ax$; $\therefore x=\dfrac{a^2-1}{2a}=\dfrac{1}{2}\left\{a-\dfrac{1}{a}\right\}$

Let $a=\left\{\dfrac{y+\sqrt{y^2+4}}{2}\right\}^3$ $\dfrac{1}{a}=\left\{\dfrac{2}{y+\sqrt{y^2+4}}.\dfrac{y-\sqrt{y^2+4}}{y-\sqrt{y^2+4}}\right\}^3=\left(\dfrac{y-\sqrt{y^2+4}}{-2}\right)^3-\dfrac{1}{a}=\left(\dfrac{y-\sqrt{y^2+4}}{2}\right)^3$

$x=\dfrac{1}{2}\left(\left\{\dfrac{y+\sqrt{y^2+4}}{2}\right\}^3+\left\{\dfrac{y-\sqrt{y^2+4}}{2}\right\}^3\right)\quad=\dfrac{1}{8}\left({}^3C_0y^3+{}^3C_2y\left(y^2+4\right)\right)=\dfrac{y^3+3y}{2}$;

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$$f^{-1}(x)=\frac{x^3+3x}{2};\qquad \sum_{n=1}^{n} f^{-1}(n)=\frac{1}{2}\left(\frac{n(n+1)}{2}\right)^2+\frac{3}{2}\frac{n(n+1)}{2}=\frac{n^2(n+1)^2}{8}+\frac{3}{4}n(n+1)$$

63. Differentiating both sides, we get

$$x^{26}(x-1)^{17}(5x-3)=\frac{1}{k}\{x^{27}.18(x-1)^{17}+(x-1)^{18}27x^{26}\}$$

$$\frac{x^{26}(x-1)^{17}}{k}((18x+27(x-1)))\quad\therefore\quad K=9$$

64. $\lim\limits_{x\to\infty}\dfrac{(1-\cos x)+(1-\cos 2x)+.....+(1-\cos 10x)}{x^2}=\dfrac{1}{2}(1^2+2^2+....+10^2)=\dfrac{385}{2}$

65. For $n>6$, $S_n=S_6+7|+8|+.....+n!$ = divi. by 7

$$\frac{S_n}{7}=\frac{873}{7}+K\ \ (K\in I)\qquad K=\frac{1}{7}\{7|+8|+.....+n!\}$$

$$\Rightarrow\left[\frac{S_n}{7}\right]=K^{+}\left[\frac{873}{7}\right]=K+124$$

$$x=S_n-7\left[\frac{S_n}{7}\right]=(S_6+7K)-7(K+124)=873-868=5$$

$$\text{Sin}^{-1}(\sin x)=\text{Sin}^{-1}(\sin 5)=5-2\pi\qquad\left(2\pi-\frac{\pi}{2}<5<2\pi+\frac{\pi}{2}\right)$$

66. $P(H)=2\times P(T)$; $P(H)+P(T)=1$

$$P(H)=\frac{2}{3}\ ;\quad P(T)=\frac{1}{3}$$

P (Head occurs atleast once) = 1 – P (No head occurrence) $=1-\dfrac{1}{9}=\dfrac{8}{9}=2\times\left(\dfrac{2}{3}\right)^2$

67. $A=\begin{bmatrix}1&0&0\\2&1&0\\3&2&1\end{bmatrix}\qquad |A|=1\qquad A^{-1}=Adj(A)=\begin{bmatrix}1&0&0\\-2&1&0\\1&-2&1\end{bmatrix}$

$$AU_1=\begin{bmatrix}1\\0\\0\end{bmatrix}\Rightarrow U_1=A^{-1}\begin{bmatrix}1\\0\\0\end{bmatrix}=\begin{bmatrix}1&0&0\\-2&1&0\\1&-2&1\end{bmatrix}\begin{bmatrix}1\\0\\0\end{bmatrix}=\begin{bmatrix}1\\-2\\1\end{bmatrix}$$

$$U_2=\begin{bmatrix}1&0&0\\-2&1&0\\1&-2&1\end{bmatrix}\begin{bmatrix}2\\3\\0\end{bmatrix}=\begin{bmatrix}2\\-1\\-4\end{bmatrix};\ U_3=\begin{bmatrix}1&0&0\\-2&1&0\\1&-2&1\end{bmatrix}\begin{bmatrix}2\\3\\1\end{bmatrix}=\begin{bmatrix}2\\-1\\-3\end{bmatrix}$$

$$U=\begin{bmatrix}1&2&2\\-2&-1&-1\\1&-4&-3\end{bmatrix}\Rightarrow |U|=3$$

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68. $I=\int_0^x \frac{x\sin^4 x}{\sin^4 x+\cos^4 x}dx=\int_0^\pi \frac{(\pi-x)\sin^4 x}{\sin^4 x+\cos^4 x}dx$

$2I=\int_0^\pi \frac{(x+x-x)\sin^4 x}{\sin^4 x+\cos^4 x}dx=x\int_0^x \frac{\sin^4 x}{\sin^4 x+\cos^4 x}dx \quad \left(f(\pi-x)=f(x)\right)$

$\Rightarrow 2I=2x\int_0^\pi \frac{\sin^4 x}{\sin^4 x+\cos^4 x}dx \Rightarrow I=x\int_0^{x/2} \frac{\cos^4 x dx}{\cos^4 x+\sin^4 x} \Rightarrow 2I=x\int_0^{x/2} dx=\frac{x^2}{2} \Rightarrow I=\frac{x^2}{4}$

69. $L_1=3x+4y+5=0$; $L_2=3x+4y+15=0$

Distance between the two lines = $\left|\frac{15-5}{\sqrt{3^2+4^2}}\right|=2$

BC = 2 units, let $\angle BAB_1=0$; AB = 2 $\cos ec\theta$, $BB_1=2\sec\theta$

$\Rightarrow ar(\Delta ABB_1)=\frac{1}{2}AB,\ BB_1=\frac{1}{2}.2\cos ec\theta.2\sec\theta=\frac{4}{\sin 2\theta}$

Now, area of parallelogram AA_1BB_1 = $2\Delta ABB_1=\frac{8}{\sin 2\theta}$

Clearly this is least when $\sin 2\theta=1 \Rightarrow \theta=\frac{\pi}{4}$

Let the slope of the line drawn be 'm' $\Rightarrow 1=\left|\frac{m+\frac{3}{4}}{1-\frac{3m}{4}}\right| \Rightarrow m=-7,\ \frac{1}{7}$

70. $(1+x)^m(1-x)^n$

$=\left({}^mC_0+{}^mC_1x+{}^mC_2x^2+....+{}^mC_mx^m\right)\left({}^nC_0-{}^nC_1x+{}^nC_2x^2+....+(-1)^n\ {}^nC_n\right)$

Coefficient of $x^2={}^mC_0{}^nC_2-{}^mC_1{}^nC_1+{}^mC_2{}^nC_0=-6$ ….(i)

Coefficient of $x={}^mC_0\left(-{}^nC_1\right)+{}^mC_1{}^nC_0=3$ …..(ii)

By (i) $\frac{n(n-1)}{2}-mn+\frac{m(m-1)}{2}=-6$ …… (iii)

By (ii) - $n+m=3$ …. (iv)

By (iii) and (iv)

$\frac{n(n-1)}{2}-(n+3)n+\frac{(n+3)(n+2)}{2}=-6$

$n(n-1)-2n(n+3)+(n+3)(n+2)=-12$

$n^2-n-2n^2-6n+n^2+5n+6=-12$

$-2n=-18 \Rightarrow n=9$

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71. $|A|=\begin{bmatrix}\alpha & \beta & \gamma\\ \alpha^2 & \beta^2 & \gamma^2\\ \beta+\gamma & \gamma+\alpha & \alpha+\beta\end{bmatrix}$ $\quad R_3 \to R_3+R_1$

$$|A|=\begin{bmatrix}\alpha & \beta & \gamma\\ \alpha^2 & \beta^2 & \gamma^2\\ \beta+\gamma & \gamma+\alpha & \alpha+\beta\end{bmatrix}=(\alpha+\beta+\gamma)\begin{vmatrix}\alpha & \beta & \gamma\\ \alpha^2 & \beta^2 & \gamma^2\\ 1 & 1 & 1\end{vmatrix}=(\alpha+\beta+\gamma)(\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)$$

$$|Adj\left(adj\left(adj\left(adj\left(A\right)\right)\right)\right)|=|A|^{2^4}=|A|^{16}=\left\{(\alpha+\beta+\gamma)\,(\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)\right\}^{16}$$

$$\frac{|Adj\left(adj\left(adj\left(adj\left(A\right)\right)\right)\right)}{\left((\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)\right)^{16}}=(\alpha+\beta+\gamma)^{16}=2^{32}.3^{16}=(12)^{16}$$

$\therefore \alpha+\beta+\gamma=12$

α, β, γ = distinct natural no.

No. of Solutions = $\quad {}^{12-1}C_{3-1}-1-{}^{3}C_2(4)$

$$={}^{11}C_2-1-12=55-13=42$$

72. $\vec{r}=\vec{c}+\lambda\vec{b}$

73. $\lim\limits_{x\to 0^+}\dfrac{\left(x-\frac{x^3}{3!}+\frac{x^5}{5!}-.....\right)+a\left(1+\frac{x}{1!}+\frac{x^2}{2!}+\frac{x^3}{3!}+.....\right)+b\left(1-\frac{x}{1!}+\frac{x^2}{2!}+\frac{x^3}{3!}+.....\right)+c\left(x-\frac{x^2}{2}+\frac{x^3}{3}-.....\right)}{x^3}$ for

finite limit, a+ b = 0

1 + a – b + c = 0; $\quad \frac{a}{2}+\frac{b}{2}-\frac{c}{2}=0\Rightarrow c=0 \qquad \therefore a+b-c=0$

74. $\frac{dy}{dx}=\frac{y\phi'(x)-y^2}{\phi(x)} \Rightarrow \frac{dy}{dx}=\frac{y\phi'(x)}{\phi(x)}-\frac{y^2}{\phi(x)}$

$$\Rightarrow\frac{1}{y^2}\frac{dy}{dx}=\frac{1}{y}\frac{\phi'(x)}{\phi(x)}-\frac{1}{\phi(x)} \Rightarrow\frac{1}{y^2}\frac{dy}{dx}-\frac{1}{y}\frac{\phi'(x)}{\phi(x)}=-\frac{1}{\phi(x)}$$

Let $\frac{1}{y}=v$

$$-\frac{1}{y^2}\frac{dy}{dx}=\frac{dv}{dx}=-\frac{dv}{dx}-v.\frac{\phi'(x)}{\phi(x)}=\frac{1}{\phi(x)}$$

$\Rightarrow \frac{dv}{dx}+v\frac{\phi'(x)}{\phi(x)}=\frac{1}{\phi(x)}$, which is linear is v.

75. (1, 1), (2, 2), (3, 3), (2, 1), (3, 2), (1, 3), (3, 1) must be added.

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